

Network Detective & Self-Healing Network

An Intelligent Network Monitoring and Recovery System

COMPUTER NETWORKS PROJECT

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The Problem

Why networks need more than passive monitoring

Traditional monitoring only detects issues

It can alert administrators that something is wrong, but it does not restore service or resolve the root cause.

Network failures can still disrupt users

- Link or device failure
- High congestion and packet loss
- High latency
- Unreachable nodes

The missing piece is action: networks need automatic diagnosis and recovery, not just alerts.

Network Detective & Self-Healing

The system continuously monitors the network and identifies abnormal behavior

Network Detective

It watches devices, links, and performance signals to spot problems early.

- Device monitoring
- Performance metrics
- Fault identification

Self-Healing Module

It responds automatically to restore connectivity and confirm recovery.

- Alternate routing
- Connection restore
- Verification

Core Concept: Monitor → Detect → Analyze → Recover → Verify

System Architecture

A continuous loop from monitoring to verified recovery.

Key technologies work together to monitor the network, detect issues, and support automated recovery.

TCP/IP	Ping	Python

Network Devices

Routers, switches, and servers form the core network layer.

How Our System Works

1

Monitor

Checks network nodes and links.

2

Detect

Flags failures, latency, and loss.

3

Diagnose

Identifies the affected component.

4

Find Path

Locates an alternate route.

5

Self-Heal

Reroutes traffic automatically.

6

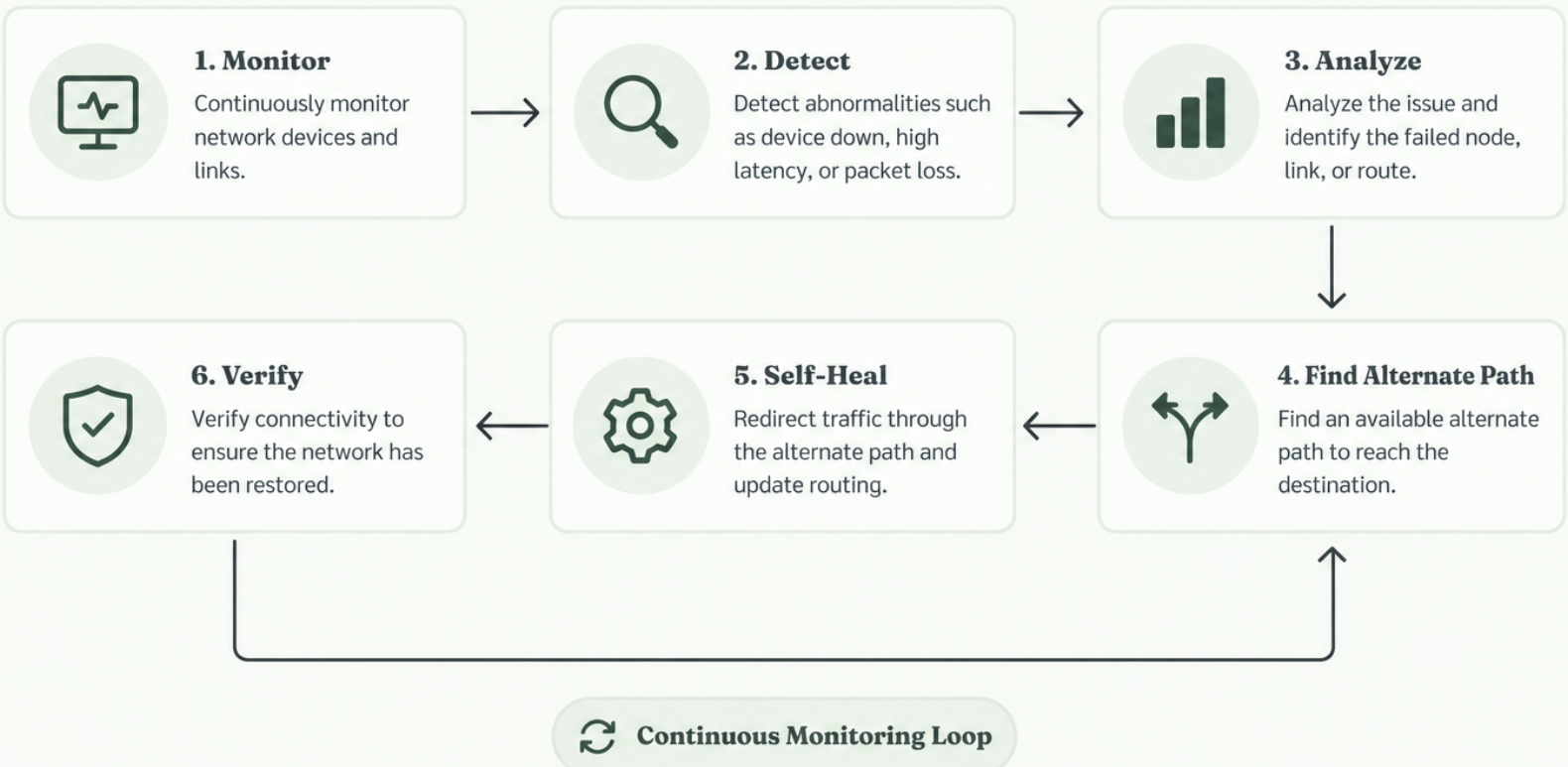
Verify

Confirms recovery.

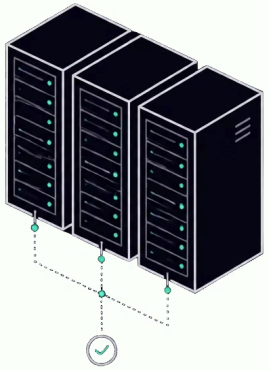
Real-World Example: If the link between B and C fails, the system detects it, finds A → B → D → C, and reroutes traffic automatically.

System Flow

The continuous flow of monitoring, detection, analysis, recovery and verification.



Advantages & Future Scope



Advantages

Automatic Failure Detection

Instantly identifies faults without waiting for manual reports.

Faster Fault Diagnosis

Pinpoints the exact failed node, link, or route in real time.

Automatic Network Recovery

Reroutes traffic and restores connectivity with no human steps.

Reduces Network Downtime

Minimizes service interruptions and improves overall reliability.

Thank You

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Computer Networks Project

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